

# Cybersecurity and Cryptography

## Principles and Practice

### Cybersecurity and Cryptography Course



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# Cryptography and Network Security

## Principles and Practice

### Time Table:

| Week                                  | Topics                                                           |
|---------------------------------------|------------------------------------------------------------------|
| <b>PART ONE: BACKGROUND</b>           |                                                                  |
| 1                                     | <b>Chapter 1:</b> Information and Network Security Concepts.     |
| 2                                     | <b>Chapter 2:</b> Introduction to Number Theory.                 |
| <b>PART TWO: SYMMETRIC CIPHERS</b>    |                                                                  |
| 3                                     | <b>Chapter 3:</b> Classical Encryption Techniques.               |
| 4                                     | <b>Chapter 4:</b> Block Ciphers and the Data Encryption Standard |
| 5                                     | <b>Chapter 5:</b> Finite Fields.                                 |
| 6                                     | <b>Chapter 6:</b> Advanced Encryption Standard.                  |
| 7                                     | <b>Chapter 7:</b> Block Cipher Operation.                        |
| 8                                     | <b>Chapter 8:</b> Random Bit Generation and Stream Ciphers.      |
| <b>PART THREE: ASYMMETRIC CIPHERS</b> |                                                                  |
| 9                                     | <b>Chapter 9:</b> Public-Key Cryptography and RSA                |
| 10                                    | <b>Chapter 10:</b> Other Public-Key Cryptosystems                |

# Cryptography and Network Security

## Principles and Practice

### Time Table:

| Week                                                      | Topics                                                                     |
|-----------------------------------------------------------|----------------------------------------------------------------------------|
| <b>PART FOUR: CRYPTOGRAPHIC DATA INTEGRITY ALGORITHMS</b> |                                                                            |
| <b>11</b>                                                 | <b>Chapter 11:</b> Cryptographic Hash Functions                            |
| <b>12</b>                                                 | <b>Chapter 12:</b> Message Authentication Codes.                           |
| <b>13</b>                                                 | <b>Chapter 13:</b> Digital Signatures.                                     |
| <b>14</b>                                                 | <b>Chapter 14:</b> Lightweight Cryptography and Post-Quantum Cryptography. |
| <b>PART FIVE: MUTUAL TRUST</b>                            |                                                                            |
| <b>15</b>                                                 | <b>Chapter 15:</b> Cryptographic Key Management and Distribution.          |
| <b>16</b>                                                 | <b>Chapter 16:</b> User Authentication.                                    |
| <b>PART SIX: NETWORK AND INTERNET SECURITY</b>            |                                                                            |
| <b>17</b>                                                 | <b>Chapter 17:</b> Transport-Level Security.                               |
| <b>18</b>                                                 | <b>Chapter 18:</b> Wireless Network Security.                              |
| <b>19</b>                                                 | <b>Chapter 19:</b> Electronic Mail Security.                               |
| <b>20</b>                                                 | <b>Chapter 20:</b> IP Security.                                            |

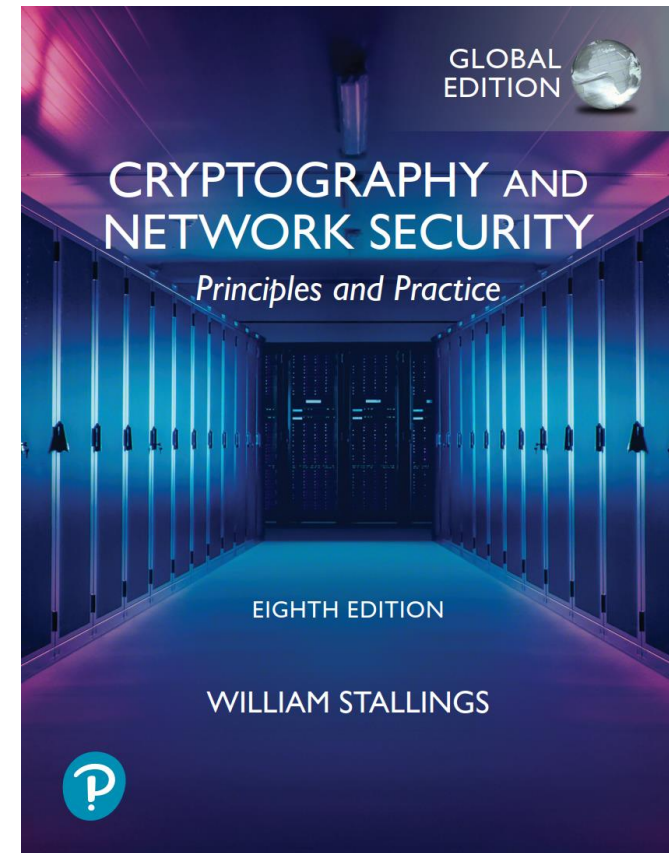
# Cryptography and Network Security

## Principles and Practice

### Chapter 3

## Classical Encryption Techniques

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## Chapter Outcomes:

### 3.1 Symmetric Cipher Model

- Cryptography
- Cryptanalysis and Brute-Force Attack

### 3.2 Substitution Techniques

- Caesar Cipher
- Monoalphabetic Ciphers
- Playfair Cipher
- Hill Cipher
- Polyalphabetic Ciphers
- One-Time Pad

### 3.3 Transposition Techniques

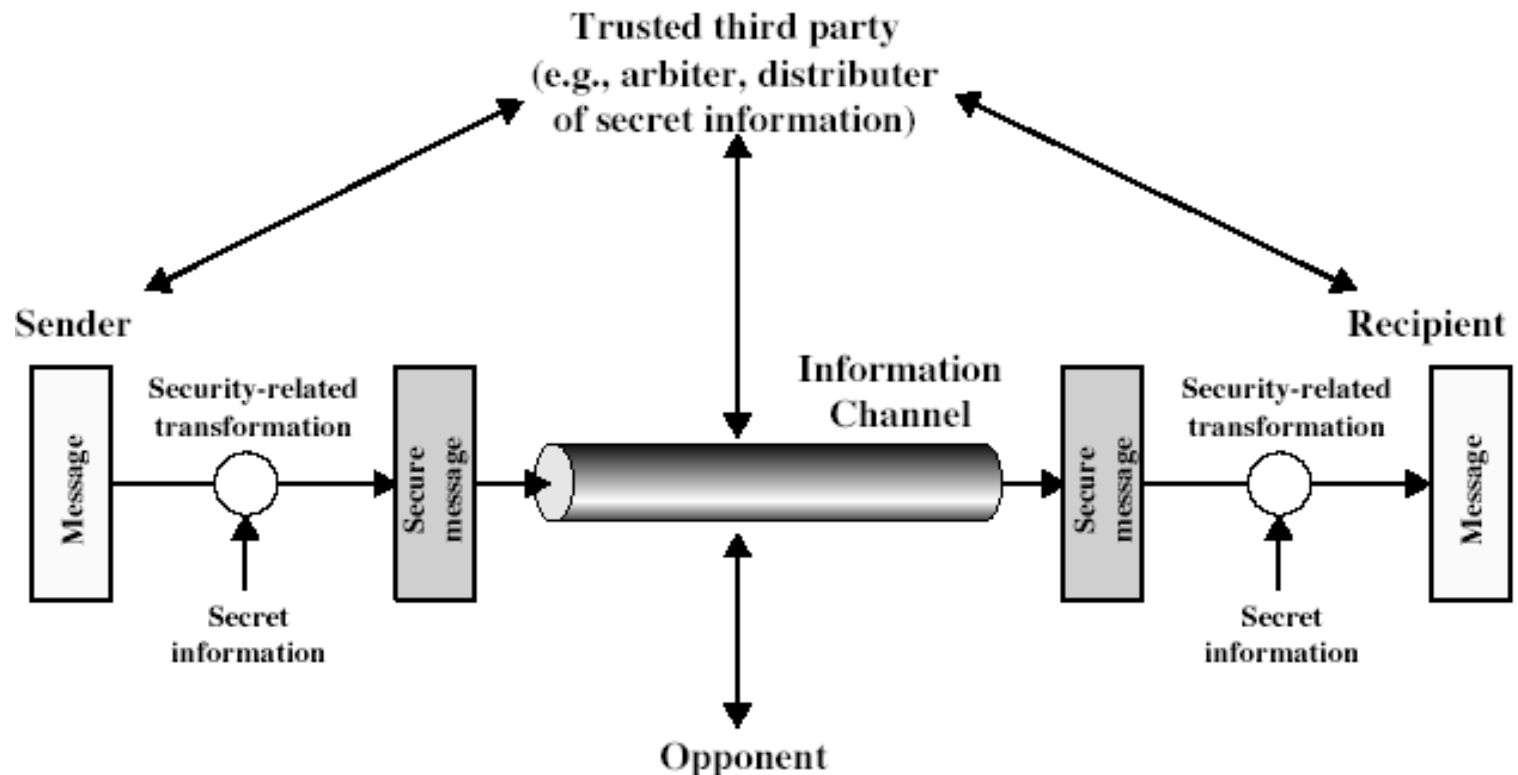
- 3.4 Key Terms, Review Questions, and Problems



# Symmetric Cipher Model

## Simplified Model of Symmetric Encryption

❑ **Symmetric encryption**, also referred to as **conventional encryption** or **single-key Encryption**.



# Symmetric Cipher Model

- **Using this model requires us to:**
  - Design a suitable algorithm for the security transformation.
  - Generate the secret information (keys) used by the algorithm.
  - Develop methods to distribute and share the secret information.
  - Specify a protocol enabling the principals to use the transformation and secret information for a security service.

# Symmetric Cipher Model (Cont.)

## Key Terms of Symmetric Cipher

- **Known as:**
  - **Conventional Encryption**
  - **Single-Key Encryption**
- **Plaintext**
  - Original text/msg
- **Ciphertext**
  - Coded msg
- **Enciphering/Encryption**
  - The process of converting the plaintext to ciphertext
- **Deciphering/Decryption**
  - The process of converting the ciphertext to plaintext

# Symmetric Cipher Model (Cont.)

- **Cryptography**
  - The developed encryption schemes
- **Cryptanalysis**
  - Techniques used to get the plaintext out of the ciphertext without prior knowledge to the encryption scheme (**breaking the code**)
- **Cryptology**
  - Both the cryptography and cryptanalysis

# Symmetric Cipher Model (Cont.)

## ❑ Cryptography

Cryptographic systems are characterized along three independent dimensions:

### 1. The type of operations used for transforming plaintext to ciphertext.

- a) **Substitution**, in which each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element,
- b) **Transposition**, in which elements in the plaintext are rearranged. The fundamental requirement is that no information be lost (i.e., that all operations are reversible).

### 2. The number of keys used.

- a) **Symmetric**, single-key, secret-key, or conventional encryption: If both sender and receiver use the same key.
- b) **Asymmetric**, two-key, or public-key encryption: If the sender and receiver use different keys

### 3. The way in which the plaintext is processed.

- a) **A block cipher**, processes the input one block of elements at a time, producing an output block for each input block.
- b) **A stream cipher**, processes the input elements continuously, producing output one element at a time, as it goes along.

# Cryptanalysis and Brute-Force Attack

- ❑ the objective of attacking an encryption system is to recover the key in use rather than simply to recover the plaintext of a single ciphertext.
- ❑ There are two general approaches to attacking a conventional encryption scheme:
  - **Cryptanalysis:** Cryptanalytic attacks rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext–ciphertext pairs.
    - This type of attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used.
  - **Brute-force attack:** The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained.
    - On average, half of all possible keys must be tried to achieve success.

# Cryptanalysis and Brute-Force Attack

**Table 3.1** Types of Attacks on Encrypted Messages

| Type of Attack    | Known to Cryptanalyst                                                                                                                                                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ciphertext Only   | <ul style="list-style-type: none"><li>■ Encryption algorithm</li><li>■ Ciphertext</li></ul>                                                                                                                                                                                                                                                             |
| Known Plaintext   | <ul style="list-style-type: none"><li>■ Encryption algorithm</li><li>■ Ciphertext</li><li>■ One or more plaintext–ciphertext pairs formed with the secret key</li></ul>                                                                                                                                                                                 |
| Chosen Plaintext  | <ul style="list-style-type: none"><li>■ Encryption algorithm</li><li>■ Ciphertext</li><li>■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key</li></ul>                                                                                                                                |
| Chosen Ciphertext | <ul style="list-style-type: none"><li>■ Encryption algorithm</li><li>■ Ciphertext</li><li>■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key</li></ul>                                                                                                                              |
| Chosen Text       | <ul style="list-style-type: none"><li>■ Encryption algorithm</li><li>■ Ciphertext</li><li>■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key</li><li>■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key</li></ul> |

# Symmetric Cipher Model (Cont.)

## ❑ Cryptanalysis

### More Definitions

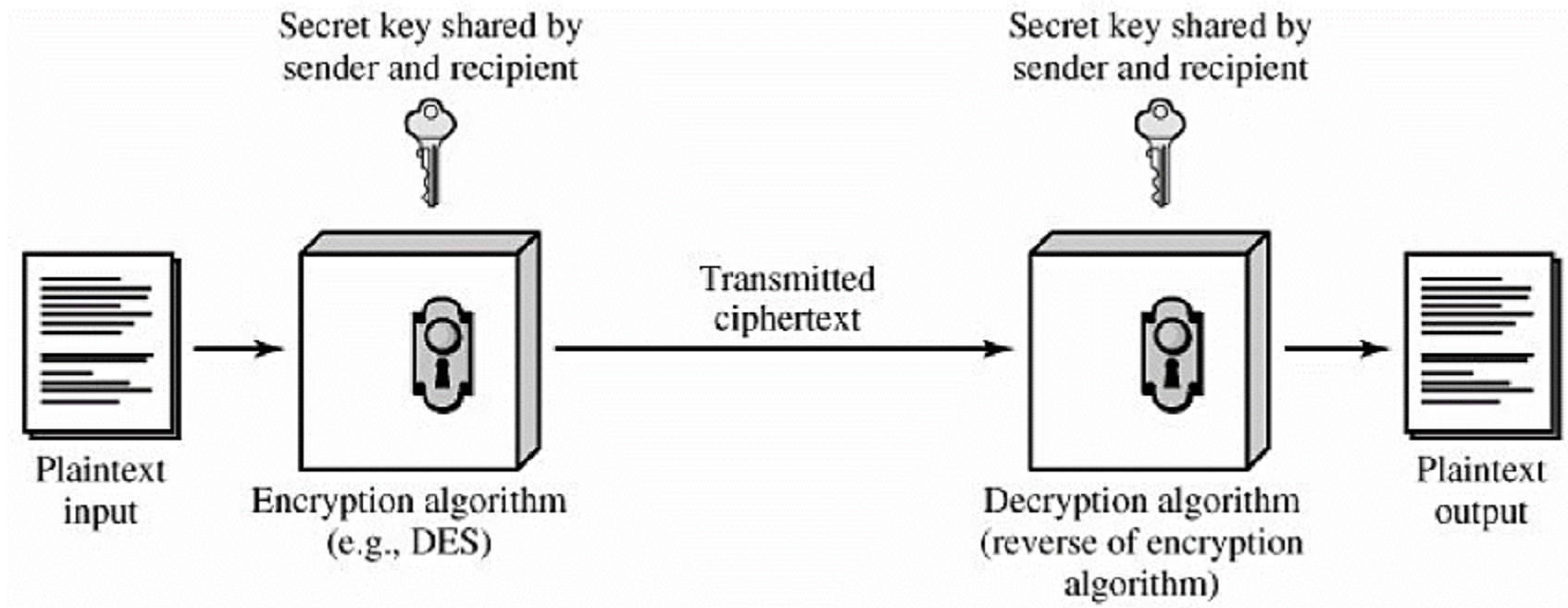
- **Unconditional Security**
  - The ciphertext provides insufficient information to uniquely determine the corresponding plaintext.
- **Computational Security**
  - The time needed for calculations is greater than age of universe

# Brute Force Search

- Always possible to simply try every key
- Most basic attack, proportional to key size

| Key size (bits)                | Number of alternative keys     | Time required at 1 decryption/ <i>ms</i>                         | Time required at $10^6$ decryption/ <i>ms</i> |
|--------------------------------|--------------------------------|------------------------------------------------------------------|-----------------------------------------------|
| 32                             | $2^{32} = 4.3 \times 10^9$     | $2^{31} \text{ ms} = 35.8 \text{ minutes}$                       | 2.15 milliseconds                             |
| 56                             | $2^{56} = 7.2 \times 10^{16}$  | $2^{55} \text{ ms} = 1142 \text{ years}$                         | 10.01 hours                                   |
| 128                            | $2^{128} = 3.4 \times 10^{38}$ | $2^{127} \text{ ms} = 5.4 \times 10^{24} \text{ years}$          | $5.4 \times 10^{18} \text{ years}$            |
| 168                            | $2^{168} = 3.7 \times 10^{50}$ | $2^{167} \text{ ms} = 5.9 \times 10^{36} \text{ years}$          | $5.9 \times 10^{30} \text{ years}$            |
| 26 characters<br>(permutation) | $26! = 4 \times 10^{26}$       | $2 \times 10^{26} \text{ ms} = 6.4 \times 10^{12} \text{ years}$ | $6.4 \times 10^6 \text{ years}$               |

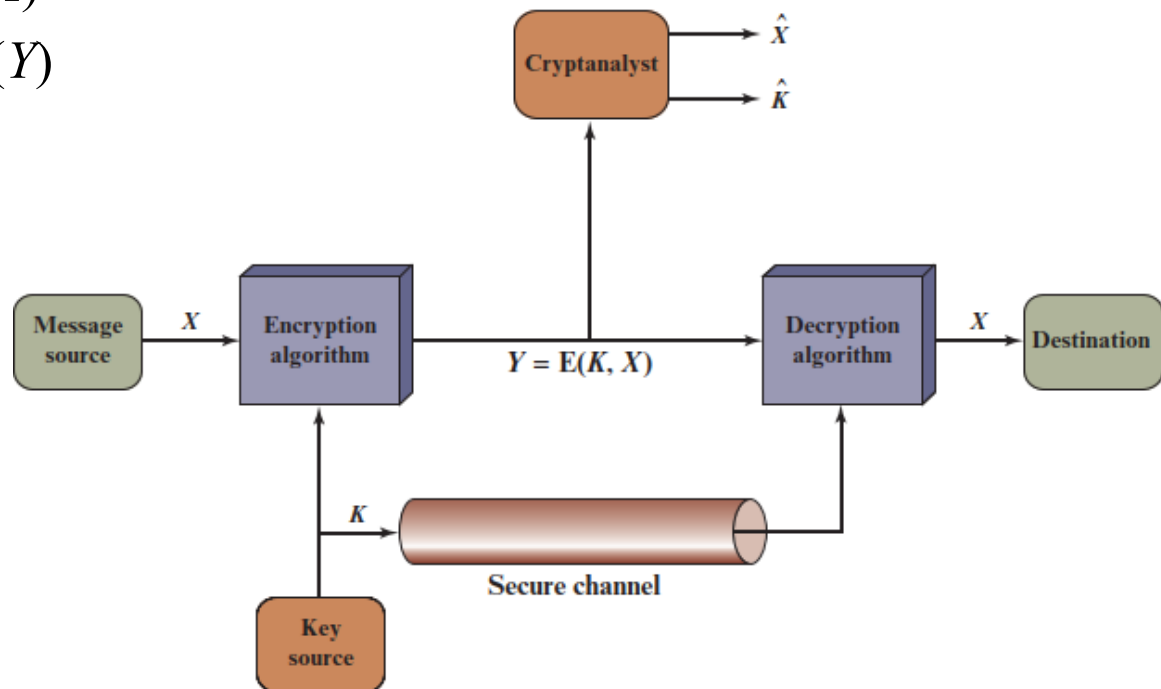
# Symmetric Cipher Model (Cont.)



# Symmetric Cipher Model (Cont.)

- **Requirements**

- Strong Key  $\rightarrow$  the opponent can not figure it out even if he/she has a number of ciphertexts
- The key must be exchanged through a secure channel
- $Y = E(K, X) \sim Y = E_K(X)$
- $X = D(K, Y) \sim X = D_K(Y)$



# Substitution Ciphers

## Lets have Fun

- You are spying on your friend Ahmed while he is chatting with John, you received the following message:

**“Ygjcxgvqmnnvjrgumfgpv”**

**Can you decrypt this message?**

# Hazards of Cyberspace

## Answer

- **Ahmed is telling John:**
  - “Ygjc xgvqmnnvjgrgumfgpv”
  - “We have to kill the president”
- **Encryption Key:**
  - **Replacement Table**
    - Plaintext ABCDEFGHIJKLMNOPQRSTUVWXYZ
    - Ciphertext CDEFGHIJKLMNOPQRSTUVWXYZAB
- **Encryption Technique**
  - Each letter is replaced by the second one after it
  - Remove blanks

# Caesar Cipher

- Earliest known substitution cipher
- by Julius Caesar
- first attested use in military affairs
- replaces each letter by 3<sup>rd</sup> one after it
- **E.g.**  
meet me after the toga party  
PHHW PH DIWHU WKH WRJD SDUWB

# Caesar Cipher (Cont.)

- Transformation :

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z  
cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- Mathematically give each letter a number

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| a | b | c | d | e | f | g | h | i | j | k  | l  | m  | n  | o  | p  | q  | r  | s  | t  | u  | v  | w  | x  | y  | z  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

- Then have Caesar cipher as:

$$C = E(p) = (p + k) \bmod (26)$$

$$p = D(C) = (C - k) \bmod (26)$$

# Caesar Cipher (Cont.)

- **Cryptanalysis**
  - Only have 26 possible ciphers
    - A maps to A,B,...Z
  - Could simply try each in turn

# Caesar Cipher (Cont.)

- **Cryptanalysis**

- Only have 26 possible ciphers
  - A maps to A,B,..Z
- Could simply try each in turn

|     | PHHW | PH | DIWHU | WKH | WRJD | SDUWB |
|-----|------|----|-------|-----|------|-------|
| KEY |      |    |       |     |      |       |
| 1   | oggv | og | chvgt | vjg | vqic | rctva |
| 2   | nffu | nf | bgufs | uif | uphb | qbsuz |
| 3   | meet | me | after | the | toga | party |
| 4   | ldds | ld | zesdq | sgd | snfz | ozqsx |
| 5   | kccr | kc | ydrp  | rfe | rmey | nyprw |
| 6   | jbbq | jb | xcqbo | qeb | qldx | mxoqv |
| 7   | iaap | ia | wbpan | pda | pkcw | lwnpu |
| 8   | hzzo | hz | vaozm | ocz | ojbv | kvmot |
| 9   | gyyn | gy | uznyl | nby | niau | julns |
| 10  | fxxm | fx | tymxk | max | mhzt | itkmr |
| 11  | ewwl | ew | sxlwj | lzw | lgys | hsjlg |
| 12  | dvvk | dv | rwkvi | kyv | kfxr | grikp |
| 13  | cuuj | cu | qvjuh | jxu | jewq | fqhjo |
| 14  | btti | bt | puitg | iwt | idvp | epgin |
| 15  | assh | as | othsf | hvs | hcuo | dofhm |
| 16  | zrrg | zr | nsgr  | gur | gbtn | cnegl |
| 17  | yqqf | yq | mrfqd | ftq | fasm | bmdfk |
| 18  | xppe | xp | lqepc | esp | ezrl | alcej |
| 19  | wood | wo | kpdob | dro | dyqk | zkbdi |
| 20  | vnnc | vn | jocna | cqn | cxpj | yjach |
| 21  | ummb | um | inbmz | bpm | bwoi | xizbg |
| 22  | tlla | tl | hmaly | aol | avnh | whyaf |
| 23  | skkz | sk | glzcx | znk | zumg | vgxze |
| 24  | rjyy | rj | fkyjw | ymj | ytlf | ufwyd |
| 25  | qiix | qi | ejxiv | xli | xske | tevx  |

# Linear Substitution Cipher

## ❑ Linear Substitution Cipher

1. It rather than just shifting the alphabet, It could shuffle (jumble) the letters arbitrarily.

❑ Then the algorithm can be expressed as follows:

- For each plaintext letter  $p$ , substitute the ciphertext letter  $C$

$$C = E(p) = (k_0 p + k_1) \bmod 26$$

# Linear Substitution Cipher

## Example:

• Find the ciphertext (c) for the following plaintext (p):  $P = \text{"Egypt"}$  where  $C = (k_0 p + k_1) \bmod 26$ . Such that:  $k_0 = 3, k_1 = 4$

## Solution:

|   |   |    |    |    |
|---|---|----|----|----|
| E | g | y  | p  | t  |
| 4 | 6 | 24 | 15 | 19 |

$$C_E = (3 \cdot 4 + 4) \bmod 26 = 16 \Rightarrow q$$

$$C_g = (3 \cdot 6 + 4) \bmod 26 = 22 \Rightarrow w$$

$$C_y = (3 \cdot 24 + 4) \bmod 26 = 76 \bmod 26 = 24 \Rightarrow y$$

$$C_p = (3 \cdot 15 + 4) \bmod 26 = 49 \bmod 26 = 23 \Rightarrow x$$

$$C_t = (3 \cdot 19 + 4) \bmod 26 = 61 \bmod 26 = 9 \Rightarrow j$$

**Then**

• The cipher text is:  $C = \text{"qw yxj"}$

## **Cryptanalysis**

- We have 26 possible keys for  $k_0$  and 26 possible keys for  $k_1$ .
- Now we need  $26 \cdot 26 = 676$  try and error to break the algorithm.

# Monoalphabetic Cipher

- ❑ A **permutation** of a finite set of elements  $S$  is an ordered sequence of all the elements of  $S$ , with each element appearing exactly once.
- ❑ For example, if  $S = \{a, b, c\}$ , there are six permutations of  $S$ :  
abc, acb, bac, bca, cab, cba
- ❑ In general, there are  **$n!$**  permutations of a set of  **$n$**  elements, because the first element can be chosen in one of  $n$  ways, the second in  $n - 1$  ways, the third in  $n - 2$  ways, and so on.

# Monoalphabetic Cipher

- Rather than just shifting the alphabet
- Could shuffle (jumble) the letters arbitrarily
- Each plaintext letter maps to a different random ciphertext letter
- The key is 26 letters long

Plain:    abcdefghijklmnopqrstuvwxyz

Cipher:  DKVQFIBJWPESCXHTMYAUOLRGZN

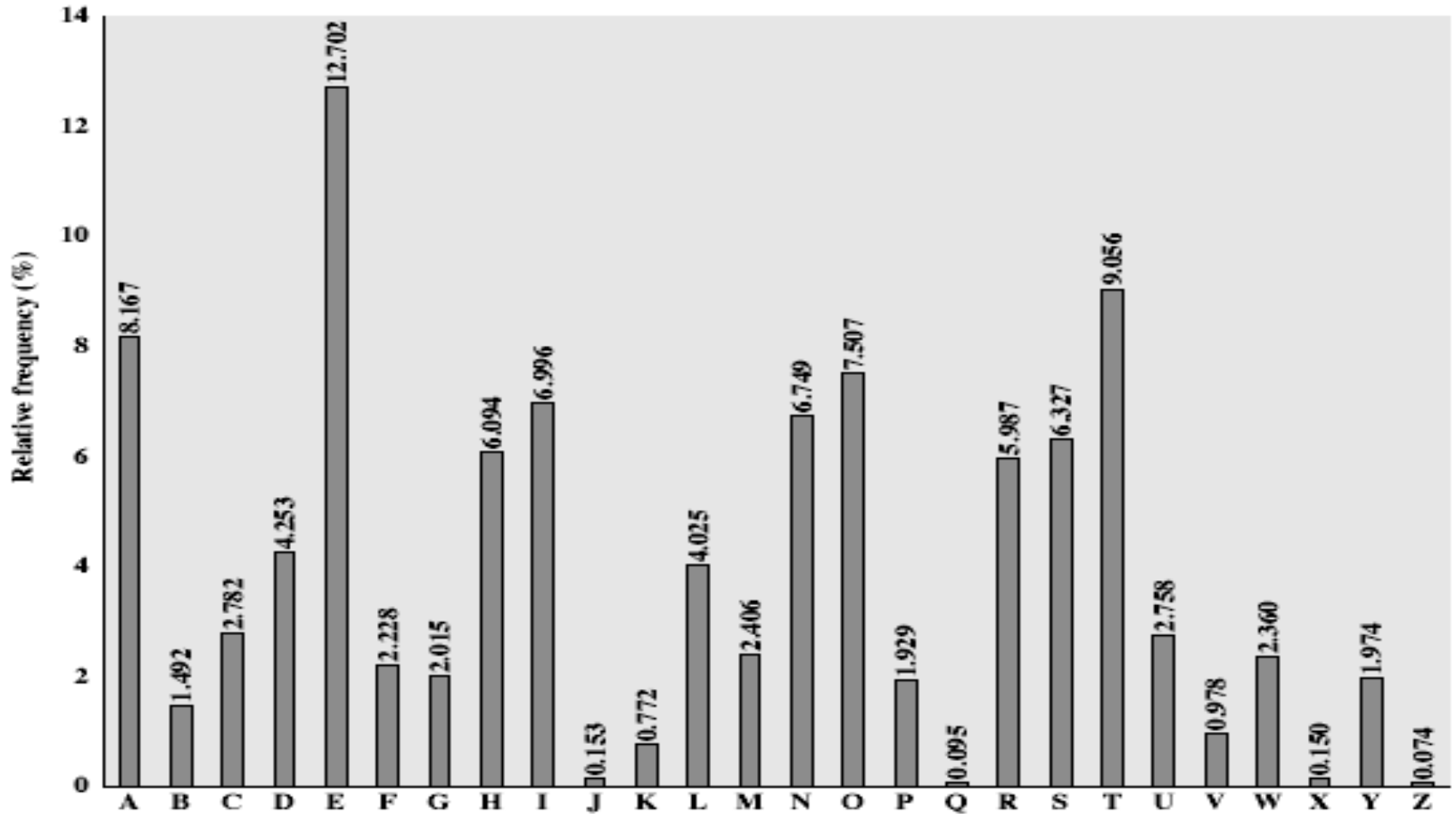
Plaintext:    ifwewishtoreplaceletters

Ciphertext:  WIRFRWAJUHYFTSDVFSFUUFYA

# Monoalphabetic Cipher Security

- now have a total of  $26! = 4 \times 10^{26}$  keys
- with so many keys, might think is secure
- but would be **!!!WRONG!!!**
- **Language Characteristics Problem**
  - Using the occurrence frequency of each letter , we can deduce the letters in the ciphertext

# English Letter Frequencies



# Playfair Cipher

- Invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair.
- Encrypts multiple letters
- Uses Playfair Matrix
- Uses some of the rules to interpret the matrix

# Playfair Key Matrix

- A 5X5 matrix of letters based on a keyword
- Fill in letters of keyword (Avoid repetition)
- Fill rest of matrix with other letters
- E.g. using the keyword MONARCHY

| M | O | N | A   | R |
|---|---|---|-----|---|
| C | H | Y | B   | D |
| E | F | G | I/J | K |
| L | P | Q | S   | T |
| U | V | W | X   | Z |

# Playfair Rules

- **Plaintext encrypted two letters at a time:**
  - **if a pair is a repeated letter, insert a filler like 'X',**
    - eg. "balloon" encrypts as "ba lx lo on"
  - **If both letters fall in the same row, replace each with letter to right (wrapping back to start from end),**
    - eg. "ar" encrypts as "RM"
  - **If both letters fall in the same column, replace each with the letter below it (again wrapping to top from bottom),**
    - eg. "mu" encrypts to "CM"
  - **Otherwise each letter is replaced by the one in its row in the column of the other letter of the pair,**
    - eg. "hs" encrypts to "BP", and "ea" to "IM" or "JM" (as desired)

# Group Activity

- Based on Playfair encryption, encrypt the word “Hello”

**Key :**

|   |   |   |     |   |
|---|---|---|-----|---|
| L | G | D | B   | A |
| Q | M | H | E   | C |
| U | R | N | I/J | F |
| X | V | S | O   | K |
| Z | Y | W | T   | P |

**Note:** The key is an arrangement of all of the alphabetic letters

# Answer

- **Step 1: Group the letters**

- He ll o
- 1<sup>st</sup> rule → repeated letters ll
- He lx lo

- **Step 2: find the corresponding text in the key**

- **He → EC** - rule 2 H and e on the same row (replace each with letter to right) → EC
- **Lx → QZ** -- rule 3 L and x at the same column (replace each with the letter below it) → QZ
- **lo → BX** -- rule 4 l and o at different rows and columns (replaced by the one in its row in the column of the other letter of the pair)

**E (Hello) “ECQZBX”**

# Security of the Playfair Cipher

- Security much improved over monoalphabetic
- Since have  $26 \times 26 = 676$  diagrams
- Was widely used for many years (eg. US & British military in WW1)
- It **can** be broken, given a few hundred letters since still has much of plaintext structure

# Polyalphabetic Ciphers

- Another approach to improving security is to use multiple cipher alphabets
- Makes cryptanalysis harder with more alphabets to guess and flatter frequency distribution
- Use a key to select which alphabet is used for each letter of the message
- Use each alphabet in turn
- Repeat from start after end of key is reached

# Vigenère Cipher

- Simplest polyalphabetic substitution cipher
- effectively multiple caesar ciphers
- key is multiple letters long  $K = k_1 k_2 \dots k_d$
- $i^{\text{th}}$  letter specifies  $i^{\text{th}}$  alphabet to use
- use each alphabet in turn
- repeat from start after  $d$  letters in message
- decryption simply works in reverse

# Vigenère Cipher

|     |   | Plaintext |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-----|---|-----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|     |   | a         | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| Key | a | A         | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|     | b | B         | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
|     | c | C         | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
|     | d | D         | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
|     | e | E         | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
|     | f | F         | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
|     | g | G         | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
|     | h | H         | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
|     | i | I         | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
|     | j | J         | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
|     | k | K         | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
|     | l | L         | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
|     | m | M         | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
|     | n | N         | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
|     | o | O         | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
|     | p | P         | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
|     | q | Q         | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
|     | r | R         | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
|     | s | S         | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
|     | t | T         | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
|     | u | U         | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
|     | v | V         | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
|     | w | W         | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
|     | x | X         | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
|     | y | Y         | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
|     | z | Z         | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |

# Example

- **eg using repeated keyword *deceptive***

key:               deceptivedeceptivedeceptive

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

- **From the previous table lookup the key letter then the plain text letter.**
- **The cipher letter is the intersection letter**

# Security of Vigenère Ciphers

- have multiple ciphertext letters for each plaintext letter
- Letter frequencies are obscured
- But not totally lost

# Autokey Cipher

- Ideally want a key as long as the message
- Vigenère proposed the **autokey** cipher
- The keyword is prefixed to message as key
- Still have frequency characteristics to attack
- **Eg. given key** *deceptive*  
key:        *deceptive*wearediscoveredsav  
plaintext: wearediscoveredsaveyourself  
ciphertext: ZICVTWQNGKZEIIGASXSTSLVVWLA

# One-Time Pad

- Select a random key that is equal to the message length.
- Use a table structure such as **Vigenère table**
- **Problems:**
  - **Generating long random keys**
  - **Bandwidth problem → sending the key as long as the Msg**

# Transposition/Permutation Ciphers

# Transposition (Cont.)

- **The letters of the message are rearranged**
- **Columnar transposition**
  - The number of columns is required
- **Example:**

THIS IS A MESSAGE TO SHOW HOW A COLMUNAR  
TRANSPOSITION WORKS

# Transposition (Cont.)

|   |   |   |   |   |
|---|---|---|---|---|
| T | H | I | S | I |
| S | A | M | E | S |
| S | A | G | E | T |
| O | S | H | O | W |
| H | O | W | A | C |
| O | L | M | U | N |
| A | R | T | R | A |
| N | S | P | O | S |
| I | T | I | O | N |
| W | O | R | K | S |

tssoh oaniw haaso lrsto imghw utpir seeoa mrook  
istwc nasna

# Group Activity

- **Given the following message**  
**“ This is the second lecture”**
- Divide the message onto a block of 5 letters block
- Transpose the message
- Use **Autokey** cipher to encrypt the result
  - Key : “ NetworkSecurity”

# Stream Vs. Block Ciphers

- Stream → converts one symbol of plaintext into a symbol of ciphertext
- Block → encrypts a group of plaintext symbols as one block.

# Recap

## ❑ Symmetric Cipher Model

- Cryptography
- Cryptanalysis and Brute-Force Attack

## ❑ Substitution Techniques

- Caesar Cipher
- Monoalphabetic Ciphers
- Playfair Cipher
- Polyalphabetic Ciphers
- One-Time Pad

## ❑ Transposition Techniques

**Thank you**



**Any Questions**